“AI-Assisted Design Pipeline for Cancer-Specific Nanocarriers”

Abstract: Despite decades of research, targeted drug delivery remains hampered by a critical issue: nanocarriers are rapidly rendered non-specific by protein corona formation (Mayordomo et al., 2025) and aptamer ligand design via systematic evolution of ligands by exponential enrichment (SELEX) is expensive. This project proposes an AI-guided design pipeline to optimize aptamer specificity and carbon quantum dot surface chemistry, addressing protein corona-driven loss of targeting, thus resolving the above. Said pipeline will fine-tune an aptamer generator by folding the output, predicting stability/binding against EGFR (used due to the existence of high-EGFR lines and known aptamers for baseline), and then using said data for supervised fine-tuning (SFT). Quantum dot (“QD”) synthesis will use classical ML to engineer syntheses from ~100 samples. The fluorescence of QDs will be used to test uptake/absorption into HEK 293 and A431 cells. The large sample space, as well as the fine-tuning, necessitates AI, as opposed to non-AI in-silico methods or SELEX. This more specific delivery may ultimately enable more precise, lower-dose drug delivery.

Motivation and Approach: The issue we are trying to solve is how to deliver payloads to cancer cells. Current cytotoxic chemotherapies (excluding antibody-drug conjugates) are non-selective in their attack on cells, which is often the limiting factor in treatment. As an example, calicheamicins are over 4000 times as effective as doxorubicin, but are rendered unusable by off-target cytotoxicity (Kumar et al., 2017). This is a trend where off target effects can limit the maximum safe dose, thus making it harder to eradicate the cancer. The state of the art for cancer drug delivery are antibody-drug conjugates

(Kumar et al., 2017), which have numerous flaws, including immunogenicity, chemical instability, and manufacturing issues. Aptamers have lower immunogenicity, higher chemical stability, and lower cost of synthesis versus monoclonal antibodies (Zhou & Rossi, 2016), and, with addition of chemical modifications (2′-F, 2′-OEt, 3′-inverted dT, 5′-PEG), can be made resistant to nucleases and perform better on stability.

However, the method of selecting an aptamer, SELEX (Systematic Evolution of Ligands by EXponential enrichment), has high labor, high cost and long time to execute, including PCR/library cost. In addition, while a SELEX experiment can explore 10^15 aptamers, there are over 4^40 ≈ 10^24 possible 40-nucleotide aptamers. The issues with SELEX thus necessitate the use of in-silico methods. Classical in-silico methods are also very costly in compute, with ViennaRNA compute being O(n3). Thus, there is a need for a pretrained aptamer generator and transformer-based folding pipeline.

Carbon quantum dots can be used as a platform for drug/aptamer delivery (Kim et al., 2019) as they are inherently fluorescent (enabling testing) and are rapidly absorbed into cells when bound. Despite this, there are issues with aggregation, specificity losses, biodistribution, pharmacokinetics, and a large sample space for synthesis. The first four issues can be attributed to protein corona. The data from quantum dots is limited, but structured, categorical, and tabular, and thus suited for gradient boosting methods. As a result, XGBoost, a gradient boosting algorithm known for high predictive accuracy (Chen & Guestrin, 2016) can be used to predict syntheses, and Bayesian optimization can use XGBoost results for synthesis design (Guo et al., 2024).

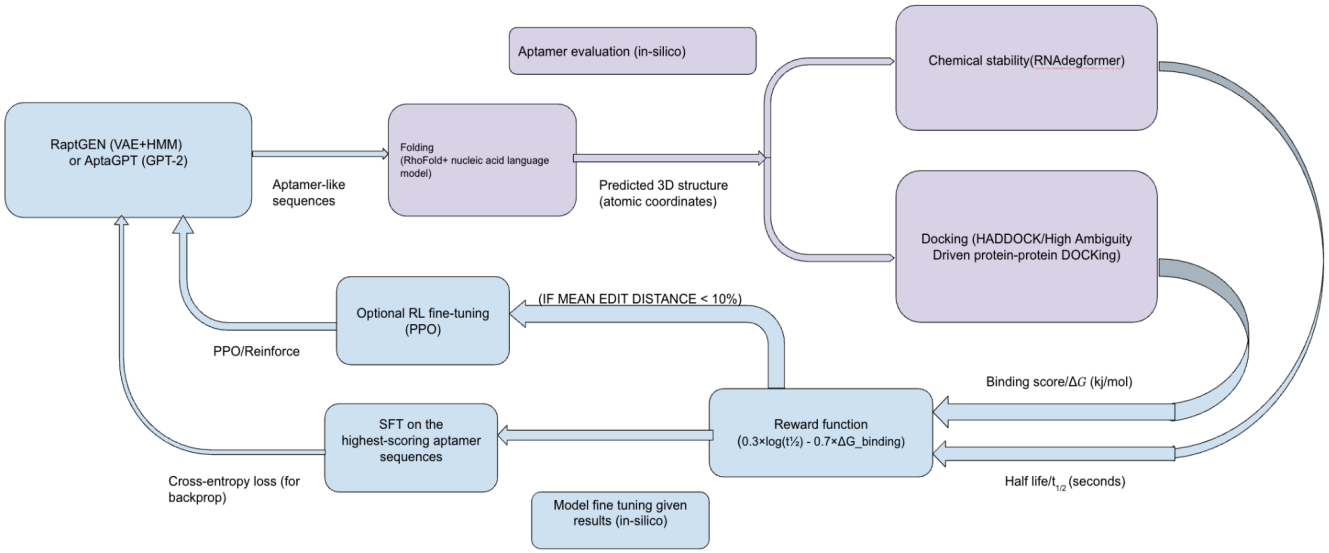
As previously mentioned, the delivery pipeline will be a CQD(Carbon Quantum Dot) with an aptamer. The goal is to use AI/ML to reduce some of the major issues and do design in-silico. The AI pipeline will involve fine-tuning an aptamer generator using a folding/docking pipeline, and training an XGboost model on QD data so Bayesian optimization can be used to generate a synthesis. This will bypass slow SELEX and synthesize aptamers that bind EGFR and quantum dots with reduced corona.

Despite the pre-existence of all models used (AptaGPT, RhoFold+, etc), no prior work has integrated AI-designed aptamers with ML-optimized quantum dots in a unified framework. Reinforcement learning/fine-tuning using a scoring function has only recently been applied to small molecules (Thomas et al., 2022); most ML models for aptamers have used supervised fine-tuning on SELEX datasets. The approach of fine-tuning on high-scoring output with diversity constraint is novel, but has been tried with GPT-Neo LLMs (Haluptzok et al., 2022).

The project is highly feasible for the budget as it's an in-silico project using existing models. The premier option for carbon quantum dot synthesis optimization is XGBoost (Guo et al., 2024). All models have been pretrained and AI-powered QD synthesis has already been attempted (Guo et al., 2024). In addition, fine tuning with structure-based scoring, which this work will expand to aptamers, has been tested (Thomas et al., 2022). Carbon quantum dots have been proposed for delivery and aptamers (IE pegaptinib) have been doing clinical trials since the early 2000s (Vinores, 2006). The approach will be split into four phases: QD design, aptamer design, synthesis and assembly, and testing.

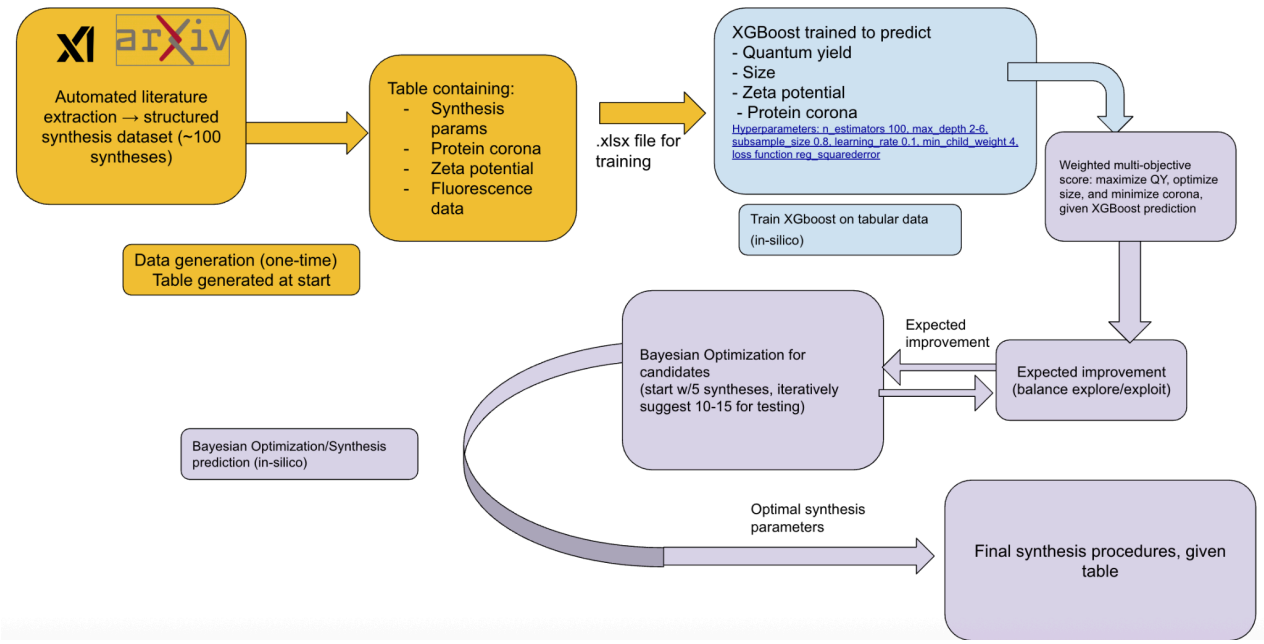
The aptamer design stems from a deep learning/fine tuning pipeline. RaptGen, a variational autoencoder w/ hidden Markov model for aptamers (Iwano et al., 2022), or AptaGPT, a fine-tuned GPT-2 transformer, will be the models used. The fine-tuning pipeline will involve folding and docking the aptamer. We plan to use RhoFold+, a nucleic acid LM, (Shen et al., 2024) for folding and Haddock (He et al., 2019) as a nucleic acid docking tool. As DNA aptamers will be used for cost and stability, we will test RhoFold+ on DNA aptamers. If prediction accuracy is poor (RMSD > 5 Å), we will either use RNA aptamers or fine-tune RhoFold on available DNA aptamers. As stability is an issue for nucleic acids, RNAdegformer (Sabnis et al., 2023) will be used to predict half life. For training, we plan to select ranked top scorers, as measured by 0.3×log(t½) - 0.7×ΔG_binding (where t½ is half life and ΔG_binding is free energy of aptamer binding, and 0.7 weight favors binding), then utilize supervised fine-tuning on the top-scoring aptamers. We may use RL fine-tuning w/ PPO if diversity begins to decline.

Aptamer Generation Pipeline



As there is limited training data for quantum dots, classical machine learning (XGboost, Bayesian Optimization) will be utilized. For training data, ~100 papers on hydrothermal synthesis of CQDs will be identified,and the synthesis parameters (N/C source, temperature, pH, additives) and results (protein corona, zeta potential, size, and PLQY/color) will be extracted manually or by LLM and then compiled into a table, which will be fed into XGBoost for prediction (Guo et al., 2024). XGBoost will be used as it can use small tabular datasets (Chen & Guestrin, 2016). Bayesian optimization will be used to design the quantum dot synthesis, via Expected Improvement acquisition function, balancing exploration (untested regions) and exploitation (high-predicted regions).

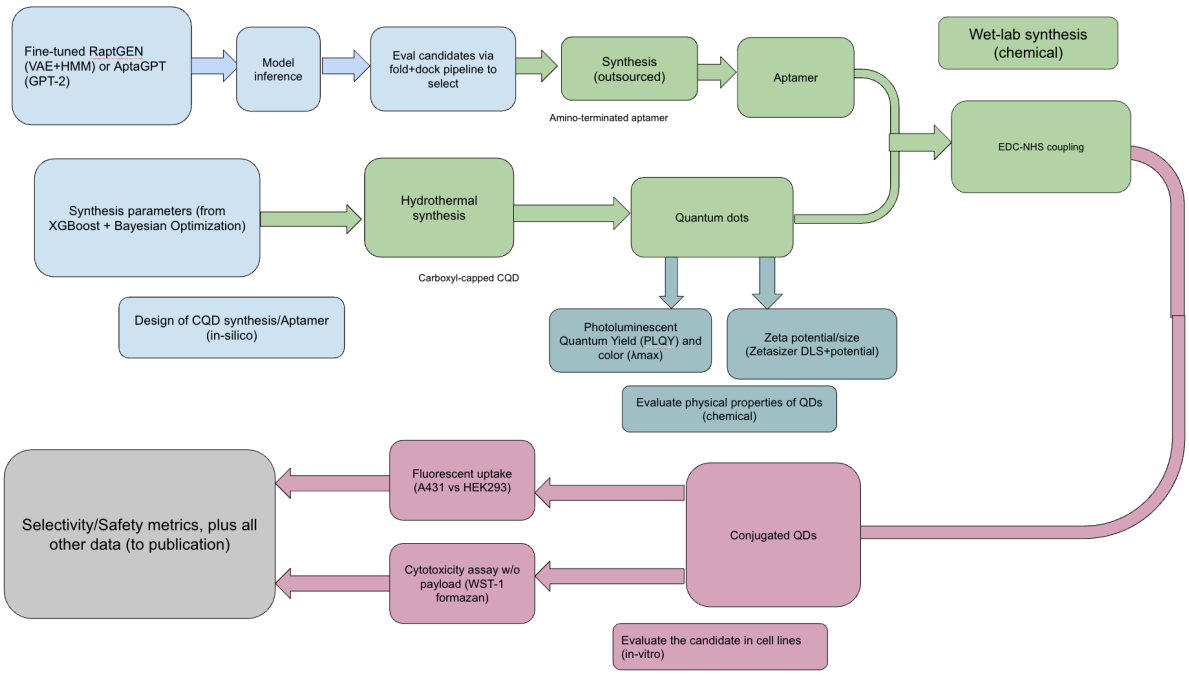
Classical ML QD Pipeline



Synthesis will then be executed. The CQD synthesis will be a hydrothermal synthesis using citric acid + nitrogen-containing precursors, the exact details of which will be optimized by the classical ML setup. For aptamers, we will utilize amino-terminated

DNA aptamers. NHS/EDC will be used for bioconjugation. After bioconjugation, there will be quantum dots with attached aptamer, which will be the final assemblies tested. We will test relative uptake into cells, plus cytotoxicity w/o payload. For cytotoxicity, we plan to use WST-1 assays to test off target cytotoxicity. Relative fluorescence increases between A431 (high EGFR) and HEK 293 (low EGFR) will be measured for uptake assessment. Quantum dot zeta potential, size, protein corona, and fluorescence (color/PLQY) will be measured to assess corona and have a baseline for tracking.

Final Experimental Workflow



Project logistics and organization: Given budget constraints, this project emphasizes proof-of-concept validation, prioritizing model-guided candidate selection with some experimental validation over in-vivo testing or directly testing with payload. Due to cost, fine tuning via SFT on the best results will be utilized, rather than extensive RLFT (used

in the event of diversity loss). The materials include synthesis precursors for the CQDs (citric acid, urea, etc), aptamers, cell lines/culture for testing, instrumentation, and especially compute time. In addition, I will need mentorship, and up to $1,000 for all materials, compute and such. I plan to acquire them via working with MIT Think.

End goals/Success criteria:

Goals	Details
Aptamers	KD w/ EGFR < 100 nM half life > 8 hours in 10% serum at 37° predicted mean free energy (MFE) less than −10 kcal/mol
Quantum dot	Protein corona limit; hydrodynamic diameter increases < 25% and zeta shift < 10 mV after 1 hour incubation in 10 percent serum Zeta potential -30 mV to -20 mV PQLY > 15% and max wavelength 550–650 nm Hydrodynamic diameter 5-20 nm
Assembly	Cytotoxicity low; reduce HEK 293 viability by less than 10 percent Uptake 5-10x higher in A431 vs HEK 293

Aptamer pipeline evaluation goals will be to generate 100 candidates. They should have ≥ 2.0 kcal/mol higher binding energy than random, mean pairwise edit distance ≥ 25–30% of sequence length, and MFE < 10 kcal/mol and half life > 8 hours. Quantum dot synthesis requires XGBoost R^2 >0.75 for Zeta potential, size, corona, and PQLY/color. At the end we will synthesize 2-3 candidate assemblies. Fluorescence-based uptake should show significantly higher fluorescence (~5-10x uptake) in cancerous cells and cytotoxicity to normal cells should be < 10 percent.

Success will be defined by stable EGFR binding aptamers (KD <100 nM), low-corona CQDs, and 5-10× selective uptake in A431 vs HEK293 with <10% normal-cell toxicity.

RISK MANAGEMENT: Risks (addressed below) include aptamers, experimental complications, overfitting, and compute. The main issue with aptamer chemistry is that they're made out of DNA, while the models used are trained on RNA. Although base-pairing is similar between DNA and RNA, which may lead to high transferability, we may train RhoFold+ on DNA. In addition, aptamer stability may be an issue. Bioconjugation and cell culture may be our second-biggest issue. Overfitting will also be handled via regularization and training. Compute cost will be handled by filtering candidates and using SFT. Finally, XGBoost, as well as having inherent advantages, will have data derived from high-quality sources, and use an 80/20 split to reduce overfits.

Category	Issue	Mitigation strategy	Fallback
Aptamer chemistry	DNA aptamers (RhoFold+ tuned on RNA)	Fine-tune RhoFold+ on DNA aptamers if RMSD > 5A	Use RLFT or synthesize RNA aptamers
	Chemical stability	Include stability as a constraint (IE, via RNADegformer) in fine-tuning	2'Fluoro modifications, add 3'-inverted dT or 5'-PEG modifications
Experimental complications	Contamination or cell death	Work w/ experienced personnel, use antibiotics in media	Access core facility cell culture services, use frozen backup
	Low EDC/NHS coupling efficiency (<50%)	Test multiple CQD:aptamer ratios optimize pH 6.0-7.1; extend reaction time	Switch to maleimide-thiol or click chemistry
Overfitting of generator	Poor generalization to new sequences	Multi-objective training; L1/L2 regularization; early stopping	Increase sampling temperature, use RLFT
	Low sequence diversity	Monitor mean pairwise edit distance (target >15%), use RLFT if too low	use RLFT if mean pairwise edit distance is too low (<15%)

Compute cost	Expensive structure prediction/docking	Pre-filter sequences before expensive steps; prioritize SFT over RLFT; use university HPC (small size = low risk)	Cloud compute backup ($250 RunPod budget); may use AdamW instead of gradient descent
Data quality	Insufficient data	LLM-assisted data extraction, 80/20 training/testing split	Use established if R^2 < 0.5

TIMELINE: The 15-week timeline is as follows:

Phase	Weeks	Activities	Deliverables
Dataset acquisition	1-3	Extract CQD synthesis data from literature into tabular format (precursors, pH, temperature, additives, time; size, zeta potential, PLQY, protein corona). Collect EGFR aptamer SELEX datasets.	Curated CQD synthesis table; aptamer training/testing datasets
Model training	3-7	Implement AI pipeline; perform inference and supervised fine-tuning on aptamers; limited RL if needed. Validate folding/docking on DNA aptamers. Train XGBoost models on CQD data.	Functional aptamer design pipeline; trained CQD property predictors
Inference/candidate selection	7-9	Generate and score aptamers; rank CQD syntheses via Bayesian optimization; analyze feature importance.	Ranked aptamer candidates; ranked CQD synthesis conditions
Synthesis/characterization	9-12	Synthesize CQDs; characterize size, zeta potential, fluorescence, protein corona. Outsource synthesis of top aptamers; assess stability and docking.	Characterized CQDs; validated aptamer candidates
Bioconjugation/cell assay	12-14	Conjugate aptamers to CQDs (NHS/EDC); confirm conjugation. Evaluate uptake (A431 vs HEK293) and cytotoxicity (WST-1).	Selectivity and cytotoxicity data
Analysis/dissemination	14-16	Analyze results; finalize manuscript; package and release code.	Final report; reproducible code repository

The main items by cost are the following, with the total under $1000:

- GPU time: $0 (university compute) -$250 (RunPod), ~200 A40/L40 hours
- 3 DNA aptamers: $40–$80, 50 nmols GRCF.
- QD precursors/consumables: $80, citric acid, urea, vials, dialysis, microplates.
- EDC/NHS reagents: $80, one small bottle of NHS and EDC for coupling.
- WST-1 viability kit (small 96 well): $150 PBIOLabs
- DLS/imaging — $150, 3–5 hr DLS + plate reader time.McGill

- Misc + shipping/contingency: $150, cell line MTA, dry ice shipping, tips, plates.

As for current work, I have identified the models needed for the pipeline and I have preliminary datasets, pipelines, and literature ready. I have not yet started training/wet-lab work, but I have completed pipeline design, model selection, and feasibility assessment. Currently, I have two CoLab notebooks demonstrating AptaGPT and XGBoost (AptaGPT has promisingly low MFE but XGBoost R^2 remains very low). From June-August 2025 I did research into CdSe/ZnS quantum dot synthesis/functionalization.

The primary barriers are more practical than conceptual, which is why I would need MIT THINK's involvement. What I don't yet have are expertise, web lab access, compute and cell lines. In addition, MIT's BSL-2 facilities and mentorship will allow me to conduct bioconjugation, cell lines, etc. Furthermore, this project involves high-risk technical decisions that require expert guidance. For example, if RNA models fail to generalize to DNA, I will need structural biology/machine learning mentorship to determine whether fine-tuning is viable or whether the project should pivot to RNA.

Thus, without MIT THINK, I can build the AI pipeline, but I cannot validate its predictions or test the final assembly without lab space, compute and expertise. MIT THINK provides the combination of interdisciplinary mentorship, BSL-2 facilities, and student autonomy to close the gap between computational design and experimental validation, allowing this project to be executed rigorously rather than speculatively.

SOURCES:

Protein Corona:

- (Mayordomo et al., 2025)
- Mayordomo, N. M., Zatarain-Beraza, A., Valerio, F., Álvarez-Méndez, V., Turegano, P., Herranz-García, L., De Aguileta, A. L., Cattani, N., Álvarez-Alonso, A., & Fanarraga, M. L. (2025). The Protein Corona Paradox: Challenges in achieving true biomimetics in nanomedicines. *Biomimetics*, *10*(5), 276. https://doi.org/10.3390/biomimetics10050276

Calicheamicin/ADC:

- (Kumar et al., 2017)
- Amit Kumar, Jason White, R. James Christie, Nazzareno Dimasi, Changshou Gao, Antibody Discovery and Protein Engineering, MedImmune, Gaithersburg, MD, United States, Platform Technologies in Drug Discovery and Validation, 4.2.4 Calicheamicins *Antibody Discovery and Protein Engineering, https://doi.org/10.1016/bs.armc.2017.08.002*

Aptamer benefits

- (Zhou & Rossi, 2016)
- Zhou, J., & Rossi, J. (2016). Aptamers as targeted therapeutics: current potential and challenges. *Nature Reviews Drug Discovery*, *16*(3), 181–202. https://doi.org/10.1038/nrd.2016.199

Aptamer + QD Combo

- (Kim et al., 2019)
- Kim, M. W., Jeong, H. Y., Kang, S. J., Jeong, I. H., Choi, M. J., You, Y. M., Im, C. S., Song, I. H., Lee, T. S., Lee, J. S., Lee, A., & Park, Y. S. (2019). Anti-EGF receptor Aptamer-Guided Co-Delivery of Anti-Cancer SIRNAs and quantum dots for theranostics of Triple-Negative breast cancer. *Theranostics*, *9*(3), 837–852. https://doi.org/10.7150/thno.30228

Classical ML (XGBoost) for QD design

- (Guo et al., 2024)
- Guo, H., Lu, Y., Lei, Z., Bao, H., Zhang, M., Wang, Z., Guan, C., Tang, B., Liu, Z., & Wang, L. (2024). Machine learning-guided realization of full-color high-quantum-yield carbon quantum dots. *Nature Communications*, *15*(1), 4843. https://doi.org/10.1038/s41467-024-49172-6

RLFT for small molecules

- (Thomas et al., 2022)
- Thomas, M., O'Boyle, N. M., Bender, A., & De Graaf, C. (2022). Augmented Hill-Climb increases reinforcement learning efficiency for language-based de novo molecule generation. *Journal of Cheminformatics*, *14*(1), 68. https://doi.org/10.1186/s13321-022-00646-z

Fine-tuning on own output

- (Haluptzok et al., 2022)

- Haluptzok, P., Bowers, M., & Kalai, A. T. (2022, July 29). *Language models can teach themselves to program better*. arXiv.org. https://arxiv.org/abs/2207.14502

Pegaptinib, an early aptamer treatment

- (Vinores, 2006)
- Vinores, S. A. (2006, September 1). *Pegaptanib in the treatment of wet, age-related macular degeneration*. https://pmc.ncbi.nlm.nih.gov/articles/PMC2426796/

RaptGen

- (Iwano et al., 2022)
- Iwano, N., Adachi, T., Aoki, K., Nakamura, Y., & Hamada, M. (2022). Generative aptamer discovery using RaptGen. *Nature Computational Science*, *2*(6), 378–386. https://doi.org/10.1038/s43588-022-00249-6

RhoFold+

- (Shen et al., 2024)
- Shen, T., Hu, Z., Sun, S., Liu, D., Wong, F., Wang, J., Chen, J., Wang, Y., Hong, L., Xiao, J., Zheng, L., Krishnamoorthi, T., King, I., Wang, S., Yin, P., Collins, J. J., & Li, Y. (2024). Accurate RNA 3D structure prediction using a language model-based deep learning approach. *Nature Methods*, *21*(12), 2287–2298. https://doi.org/10.1038/s41592-024-02487-0

Haddock

- (He at al., 2019)
- Jiahua He, Jun Wang, Huanyu Tao, Yi Xiao, Sheng-You Huang, HNADOCK: a nucleic acid docking server for modeling RNA/DNA–RNA/DNA 3D complex structures, Nucleic Acids Research, Volume 47, Issue W1, 02 July 2019, Pages W35–W42, https://doi.org/10.1093/nar/gkz412

RNADegFormer

- (Sabnis et al., 2023)
- Shujun He , Baizhen Gao , Rushant Sabnis , Qing Sun, RNAdegformer: Accurate prediction of mrna degradation at nucleotide resolution with Deep Learning | Briefings in Bioinformatics | Oxford academic. (2023). https://academic.oup.com/bib/article/24/1/bbac581/6986359

XgBoost algorithm:

- (Chen & Guestrin, 2016)
- Chen, T., & Guestrin, C. (2016). XGBoost: a Scalable Tree Boosting System. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining - KDD '16, 1(1), 785–794. https://doi.org/10.1145/2939672.2939785